

Bill of Materials (BOM) SESI ChangeMakers - Americana, SP											
Item	Categoria	Component Name/Description	Quantity (units)	Reference/Model	Supplier	Status	Kit/Customized	Unit Price (US\$)	Subtotal (US\$)	Supporter	Notes
1	Mechanical Structure	3D-Printed Chassis	3	Custom	In-House Manufacturing	New	Customized			SESI - SP	Main Robot Structure and Component Mounting Support
2	Motorization	DC Geared Motor	4	25GA370 Motor	NEWPORT ROBOTICA	New	Kit	\$16,96	\$67,84	SESI - SP	Responsible for Robot Movement
3	Motorization	50 mm Omni Wheel	4	In-House Manufactured Wheels	In-House Manufacturing	New	Customized			SESI - SP	Enables Omnidirectional Movement
4	Main Control	Esp32 S3 DevKit	4	Wi-Fi/Bluetooth Module	AUTOCORE	New	Kit	\$17,98	\$71,92	SESI - SP	Main Robot Microcontrollers
5	Motor Control	Motor Driver	2	TB6612FNG Motor Drivers	AUTOCORE	New	Kit	\$3,58	\$7,16	SESI - SP	Motor Speed and Direction Control
6	Ball Sensors	TSP02240	12	Vishay TSSP4038 IR Receiver	ROBOCORE TECNOLOGIA	New	Kit	\$2,38	\$28,56	SESI - SP	Mounted Around the Edge of the Circular Board (IR Seeker)
7	Line Sensors	16-Channel Multiplexer	2	CD74HC4067	AUTOCORE	New	Kit	\$1,38	\$2,76	SESI - SP	LDR Multiplexing
8	Field Sensors	Digital Compass	1	HMC5883L Compass Module	AUTOCORE	New	Kit	\$4,98	\$4,98	SESI - SP	For Field Alignment
9	Field Sensors	HC-SR04 Ultrasonic Sensor	4	HC-SR04	ROBOCORE TECNOLOGIA	New	Kit	\$1,74	\$6,96	SESI - SP	Front, Back, Left, Right
10	Computer Vision	Camera with Microcontroller	1	OpenMV Cam RT1062	OpenMV	New	Kit	\$398,18	\$398,18	SESI - SP	Visual Processing and Object Detection
11	Interface	OLED Display	1	0.96" I2C	ROBOCORE TECNOLOGIA	New	Kit	\$4,75	\$23,00	SESI - SP	For Debugging
12	Interface	On/Off Switch	1	-	FORTELETRO	New	Kit	\$0,60	\$3,00	SESI - SP	Used as the Main Power Switch
13	Interface	Digital Voltmeter	1	3-Digit 4.5-30 VDC	ROBOCORE TECNOLOGIA	New	Kit	\$1,71	\$8,54	SESI - SP	Overall Battery Voltage Monitoring
14	Custom PCB	Upper Circular PCB	1	EasyEDA/Custom	TEC CI TECNOLOGIA	New	Customized	\$31,37	\$31,37	SESI - SP	Structural Support and Electronic Integration
15	Custom PCB	Main Circular PCB	1	EasyEDA/Custom	TEC CI TECNOLOGIA	New	Customized	\$36,10	\$36,10	SESI - SP	Main Signal Distribution
16	Custom PCB	Kicker Circular PCB	1	EasyEDA/Custom	TEC CI TECNOLOGIA	New	Customized	\$29,73	\$29,73	SESI - SP	Electronic Kicking Mechanism
17	Custom PCB	Lower Circular PCB	1	EasyEDA/Custom	TEC CI TECNOLOGIA	New	Customized	\$38,25	\$38,25	SESI - SP	Structural and Electrical Base of the System
18	Line Sensors	LDR Photoresistor Sensor	32	5MM	RYNDACK COMERCIO	New	Kit	\$0,08	\$2,56	SESI - SP	White Field Line Detection
19	Power Supply	Voltage Regulator Module	1	Mini 560 Step-Down Converter	ROBOCORE TECNOLOGIA	New	Kit	\$1,56	\$8,50	SESI - SP	Provides Power to the Main Board
20	Power Supply	Step-Up Converter Module	1	XL6009 Boost Converter	AUTOCORE	New	Kit	\$1,78	\$1,78	SESI - SP	Boosts Voltage for the Kicker Board
21	Power Supply	2200 µF Capacitor	2	2200 µF Capacitor	AUTOCORE	New	Kit	\$1,38	\$2,76	SESI - SP	Voltage Filtering and Stabilization
22	Power Supply	2.2 µF Electrolytic Capacitor		2.2 µF 16V Capacitor	AUTOCORE	New	Kit	\$0,05		SESI - SP	Reduction of Power Supply Fluctuations and Noise
23	Power Supply	1000 µF Capacitor	1	1000 µF 63V Capacitor	NEWPORT ROBOTICA	New	Kit	\$0,74	\$0,74	SESI - SP	Auxiliary Power Supply Filtering
24	Lighting Sensor	Red LEDs	32	1.8-2.2V	RYNDACK COMERCIO	New	Kit	\$0,02	\$0,77	SESI - SP	Controls Surface Illumination
25	Power Supply	LiPo Battery	1	11.1 v 2200 mah	NEWPORT ROBOTICA	New	Kit	\$67,97	\$67,97	SESI - SP	Main Power Source for the Robot
26	Main Control	PC817 Optocoupler	1	Optocoupler	NEWPORT ROBOTICA	New	Kit	\$0,20	\$1,00	SESI - SP	Protects the Control Electronics from Noise and Voltage Spikes
27	Computer Vision	Convex Mirror	1	Custom	In-House Manufacturing	New	Customized			SESI - SP	Expands the Camera's Field of View
28	Interface	Tactile Switch	3	(Push-Button) 6X6X5,0MM	ROBOCORE TECNOLOGIA	New	Kit	\$0,31	\$0,94	SESI - SP	Quick System Function Activation
29	Motorization	Motor Bell Housing	4	4 mm Shaft Coupler	ROBOCORE TECNOLOGIA	New	Kit	\$3,94	\$15,77	SESI - SP	Mechanical Coupling Between Wheel and Motor
30	Solenoid Control	IRL244N MOSFET	1	IRL244N	AUTOCORE	New	Kit	\$1,58	\$1,58	SESI - SP	Electronic Switching of the Power System
31	Interface	KRE Connector	6	2-Position Terminal Block, 5 mm Pitch	NEWPORT ROBOTICA	New	Kit	\$0,15	\$0,90	SESI - SP	Facilitates Removable Electronic Connections
32	Power Supply	Schottky Diode	3	1N5819 40V 1A	RYNDACK COMERCIO	New	Kit	\$0,05	\$0,14	SESI - SP	Reverse Current Protection
33	Interface	4-Pin Female KK Connector	4	2.5 mm 4-Pin Female Connector	RYNDACK COMERCIO	New	Kit	\$0,04	\$0,15	SESI - SP	Communication Between Sensors and Boards
34	Main Control	220 Ω Resistor	32	220 Ω 1/4-Watt Resistor	RYNDACK COMERCIO	New	Kit	\$0,01	\$0,32	SESI - SP	Electrical Current Limiting
35	Main Control	10 Ω Resistor	32	10 Ω 1/4-Watt Resistor	RYNDACK COMERCIO	New	Kit	\$0,01	\$0,32	SESI - SP	Electronic Signal Conditioning and Protection
36	Interface	2-Pin Male KK Connector	2	2.54 mm 180° Male Header	RYNDACK COMERCIO	New	Kit	\$0,04	\$0,07	SESI - SP	Power and Signal Connection
37	Interface	2-Pin Female KK Connector	2	2.5 mm 2-Pin Female Connector	RYNDACK COMERCIO	New	Kit	\$0,02	\$0,04	SESI - SP	Removable Interface Between Modules
38	Interface	Limit Switch	1	Micro Switch	RYNDACK COMERCIO	New	Kit	\$0,53	\$0,53	SESI - SP	Mechanical Position Detection
39	Main Control	Carbon Film Resistor	1	100 Ω 1/4-Watt Resistor	RYNDACK COMERCIO	New	Kit	\$0,01	\$0,01	SESI - SP	Voltage Control and Division
40	Actuator	12V Mini Solenoid Electric Lock	1	12V	RYNDACK COMERCIO	New	Kit	\$8,82	\$8,82	SESI - SP	Electromagnetic Mechanical Actuation
41	Main Control	BC327 Transistor	1	BC327 Transistor	RYNDACK COMERCIO	New	Kit	\$0,04	\$0,04	SESI - SP	Electronic Amplification and Switching
42	Main Control	BC337 Transistor	1	BC337 Transistor	RYNDACK COMERCIO	New	Kit	\$0,06	\$0,06	SESI - SP	Signal and Actuation Control
43	Main Control	100 kΩ Resistor	40	100 kΩ Resistor × 1/4 W	RYNDACK COMERCIO	New	Kit	\$0,01	\$0,48	SESI - SP	Pull-Up and Logic Stabilization
44	Power Supply	Ceramic Disc Capacitor	20	100 nF 50 V Capacitor	AUTOCORE	New	Kit	\$0,02	\$0,40	SESI - SP	High-Frequency Noise Filtering
45	Power Supply	Electrolytic Capacitor	7	470 µF 25 V Capacitor	AUTOCORE	New	Kit	\$0,07	\$0,49	SESI - SP	Power Supply Stabilization
46	Power Supply	20A 60V Rectifier Diode	1	20A 60V	AUTOCORE	New	Kit	\$1,38	\$1,38	SESI - SP	Rectification and Overcurrent Protection
47	Mechanical Structure	Acrylic Tube	1	Ø60 × 2 mm Wall Thickness × 1 m	D.K. FREITAS GALLINARI	New	Kit	\$63,00	\$63,00	SESI - SP	Structural Protection and Finishing
48	Interface	4-Pin Male KK Connector	4	2.5 mm Male Connector, 4-Pin	RYNDACK COMERCIO	New	Kit	\$0,06	\$0,24	SESI - SP	Communication Between Sensors and Boards
49	Power Supply	5D-9 NTC Thermistor (5 Ω)	1	5 Ω NTC Thermistor, 9 mm	FORTELETRO	New	Kit	\$0,74	\$3,70	SESI - SP	Thermistor for Main Power Supply
Total Cost per Robot:									\$943,82		
Total Project Cost:									\$1.887,65		